Gleipnir: Frontier-Quality AI Control Monitors at Small-Model Latency

Two-week sprint proposal | Jeremias Ferrao | MATS exploration phase | 19 October 2026

1 The project

Gleipnir trains small open-weight monitors that score AI-agent trajectories for scheming, sabotage, deception and other control-relevant behaviour. The core recipe is *binary-logit distillation*: a strong teacher reads the trajectory and its probability on the decision token (`0|1`) becomes a soft target for a LoRA student that emits the same single token. There is no generated rationale, so a monitor call costs one prefill. On the six-source, 6,395-row OOD suite of Sinha et al. (mean pAUROC@20, never used for selection), the released **Gleipnir 4B** scores **0.782 at $0.26 per 1K** evaluations, the cheapest point on the cost–performance frontier, ahead of the paper's trained Qwen3.5-4B SFT+RL (0.684) and Qwen3.5-27B SFT+RL (0.831 at $2.78). Trained the same way, **Gleipnir 27B** (Qwen3.8-27B) reaches **0.935**: above its own teacher Kimi K3 (0.908 at $25) and close to prompted Claude Opus 4.6 (0.942 at $95), at about 1/15 of Kimi's cost.

2 Sprint plan (deliverable: Gleipnir v1 model release + technical report)

Week 1: make the 4B much better, using Gleipnir 27B as an intermediate teacher.

- **Teacher-assistant distillation** (Kimi K3 → Gleipnir 27B → 4B). The 27B shares the 4B's exact 248K-token vocabulary and model family, is now a stronger monitor than Kimi, and is cheap enough to relabel freely. The unadapted 27B was a *worse* teacher than Kimi (−1.9 ID points), so the gain must come from the trained intermediate.
- **Richer supervision channels.** (i) Full-vocabulary KL at the decision position. A first probe diagnostic finds the non-decision tail adds little over the binary margin (+0.04 ID / +0.7 OOD points), so this gets one controlled arm rather than a sweep. (ii) Soft auxiliary targets such as the *index of the harmful action*, giving the student grounded evidence about *where* the misbehaviour is, which also helps human auditors.
- Recipe and checkpoint choices are frozen on the held-out ID set; OOD is read once, at the end.

Week 2: find the blind spots and evaluate on benchmarks the project has never touched.

- **Adversarial attacks on the final monitors.** Because the score is a single logit margin, white-box attacks are straightforward: gradient-guided token optimisation (GCG-style) restricted to text an agent controls (its messages, comments and tool arguments), with embedding-space attacks as an upper bound. Then test whether the attacks transfer to black-box settings, to the 27B and to frontier monitors. Report attack success against edit budget and categorise the blind spots found.
- **Fresh benchmarks.** One-shot evaluation of the frozen release checkpoints on MonitoringBench and SleightBench (unseen throughout the project), with metrics and baselines fixed before scoring.
- **Release.** Model cards, weights, evaluation code and the technical report.

3 Where Neel's input would help most

- **Attack threat model.** White-box gradient attacks are the easiest to run, but a scheming agent will not have gradients. Which attacker (white-box, transfer, LLM-rewriter, latent) would give results the control community finds convincing, and is a white-box blind-spot map still valuable as a worst-case bound?
- **New benchmarks.** Access to or contacts for MonitoringBench and SleightBench, and suggestions for other held-out monitoring evaluations that would make the release more credible.
- **Interpretability and scope.** Which mechanistic analyses of the monitor (e.g. what drives its suspicion score) would most strengthen the report, and what to cut if two weeks proves tight.

4 The rest of the suite (described in the final report and release)

- **Inference optimisation.** Prefill-only B200 serving with FP4/MXFP8 kernels and prefix caching; the quantised 27B reaches 2.08× throughput for −0.17 ID points, and a 1K–1M-token long-context sweep.
- **Quantisation-aware distillation (QAD)** so that low-precision deployment copies keep BF16 ranking quality.
- **Mechanistic ablations against injection-triggered suspicion.** Monitors flag harmless code once monitor-directed injection or jailbreak text is present; we ablate the responsible directions in training and serving to remove that shortcut without losing harmful-action recall.